

Assignment 1

1. Write a program that prompts the user to enter an integer and then prints it.

Example:

Input:

Enter an integer: 5

Output:

You entered: 5

2. Write a program that asks the user to input three integers and then prints their sum.

Example:

Input:

Please enter the first number: 5

Please enter the second number: 3

Please enter the third number: 4

Output:

The sum of the three integers is: 12

3. Write a program that asks the user to input three integers (only whole numbers, no floating-point values i.e. neither float nor double) and then prints their average as a decimal number.

Example:

Input:

Please enter the first number: 6

Please enter the second number: 4

Please enter the third number: 4

Output:

The average is: 4.66667

4. Write a program that calculates the perimeter and area of a rectangle. The program should prompt the user to enter the width and height of the rectangle.

Equations:

- Perimeter = $2 \times (\text{Width} + \text{Height})$

- Area = $\text{Width} \times \text{Height}$

Example:

Input:

Enter the width: 5

Enter the height: 3

Output:

The perimeter of the rectangle is: 16

The area of the rectangle is: 15

5. Write a program that calculates the sum of the digits of a three-digit number. The user should enter a single three-digit number, and the program should compute the sum of its digits.

Example:

Input:

Enter a three-digit number: 132

Output:

The sum of the digits is: 6

Example:

Input:

Enter a three-digit number: 451

Output:

The sum of the digits is: 10

6. Write a program that converts a given integer (representing seconds) into hours, minutes, and seconds.

Example:

Input:

Enter the time in seconds: 25300

Output:

The equivalent time is: 7 hours, 1 minute, and 40 seconds

7. Write a program that prompts the user to enter an integer. The program should print "negative" if the number is negative, "zero" if the number is 0, and "positive" otherwise.

Example:

Input:

Enter a number: -5

Output:

negative

Example 2:

Input:

Enter a number: 10

Output:

positive

Example 3:

Input:

Enter a number: 0

Output:

zero

8. Write a program that reads a digit (from 1 to 5) and displays it as a word.

Example:

Input:

Enter a digit: 4

Output:

four

9. Write a program that reads five numbers and calculates the sum of all the odd values among them.

Example:

Input:

Enter five numbers: 11 17 13 12 5

Output:

Sum of all odd values: 46

10. Write a program that reads a year and checks whether it is a leap year.

Note: - A leap year is divisible by 400 or by 4 only (but not by 100 unless also divisible by 400).

A year divisible by 400 →  2000 is a leap year

A year divisible by 4 and 100 but not by 400 →  1900 is not a leap year

A year divisible by 4 but not 100 →  2024 is a leap year

A year not divisible by 4 →  2023 is not a leap year

Example 1:

Input:

Enter a year: 2024

Output:

2024 is a leap year

Example 2:

Input:

Enter a year: 1900

Output:

1900 is not a leap year

11. Write a program that takes the user's birth year as input and determines if they are 18 years old or older.

Note: - Given that we calculate according to the current actual year (2025)

Example:

Input:

Enter your Birth year: 2007

Output:

You are 18 years or older

Example:

Input:

Enter your Birth year: 2006

Output:

You are 18 years or older

Example:

Input:

Enter your Birth year: 2008

Output:

You are younger than 18 years

12. A school has the following grading system based on marks:

- Below 25 → Grade: F
- 25 to 45 → Grade: E
- 45 to 50 → Grade: D
- 50 to 60 → Grade: C
- 60 to 80 → Grade: B
- 80 or above → Grade: A

Write a program that asks the user to enter their marks and prints the corresponding grade.

Example 1:

Input:

Enter your marks: 20

Output:

Your grade is: F

Example 2:

Input:

Enter your marks: 25

Output:

Your grade is: E

Example 4:

Input:

Enter your marks: 55

Output:

Your grade is: C

Example 3:

Input:

Enter your marks: 90

Output:

Your grade is: A

13. Search for the Switch case statement and use it to write a program that reads an integer between 1 and 12 and prints the corresponding month name.

Note: - You must solve this question using the Switch case.

Example:

Input:

Insert a number between 1 to 12: 8

Output:

August

14. Search for the ternary operator and use it to write a program that checks whether a number is even or odd.

Note: - You must solve this question using the Ternary operator.

Example:

Input:

Enter a number: -8

Output:

even

Example 2:

Input:

Enter a number: -7

Output:

odd

15. Write a program that takes a single character from the user and checks whether it is a vowel (a, e, i, o, u) or a consonant.

Note: - You must handle the uppercase letters too. like A, B

Example:

Input:

Enter a character: o

Output:

vowel

Example 2:

Input:

Enter a character: b

Output:

consonant

Example 3:

Input:

Enter a character: A

Output:

vowel

Example 4:

Input:

Enter a character: B

Output:

consonant

16. Write a program that takes three integers from the user and prints the maximum and minimum values.

Note: Do not use built-in functions like max or min.

Example 1: Distinct numbers

Input:

Enter three numbers: -7 -8 -5

Output:

Max element = -5

Min element = -8

Example 2: All numbers are the same

Input:

Enter three numbers: 4 4 4

Output:

Max element = 4

Min element = 4

Example 3: Two numbers are the same

Input:

Enter three numbers: 7 7 3

Output:

Max element = 7

Min element = 3

Example 4: Mixed positive and negative values

Input:

Enter three numbers: -2 10 -7

Output:

Max element = 10

Min element = -7